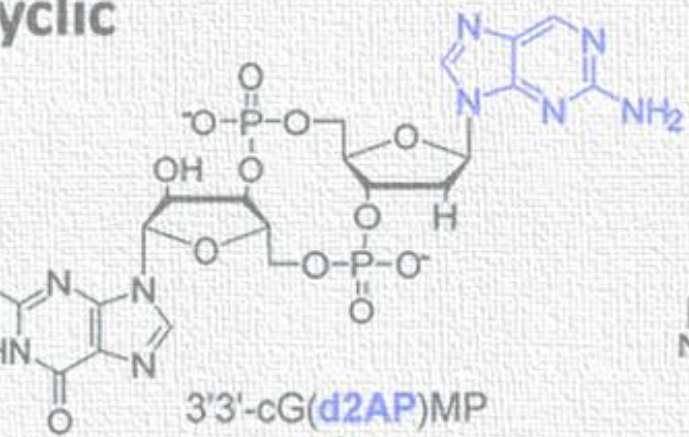


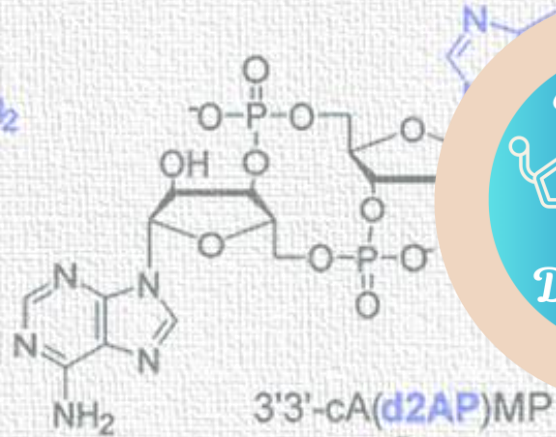
yclic



3'3'-cG(**d2AP**)MP

1

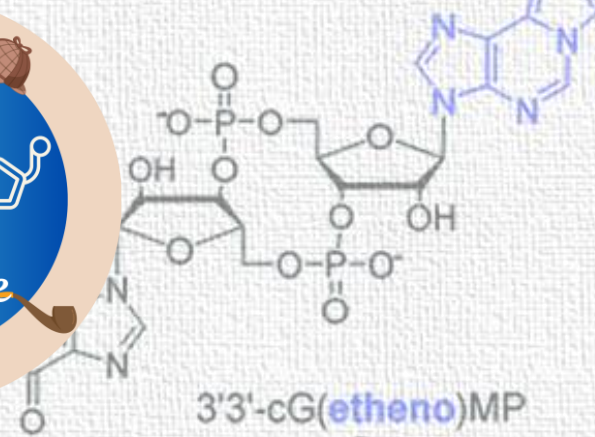
ex: 310 nm
em: 375 nm



3'3'-cA(**d2AP**)MP

2

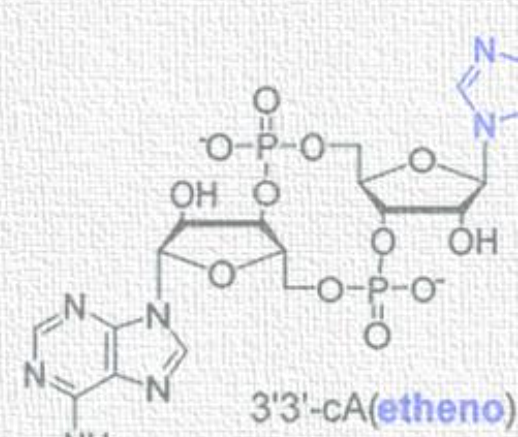
ex: 310 nm
em: 375 nm



3'3'-cG(**etheno**)MP

3

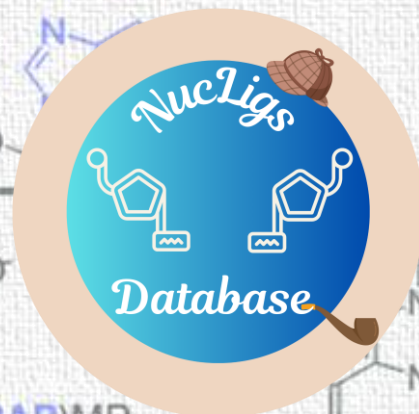
ex: 345 nm
em: 435 nm



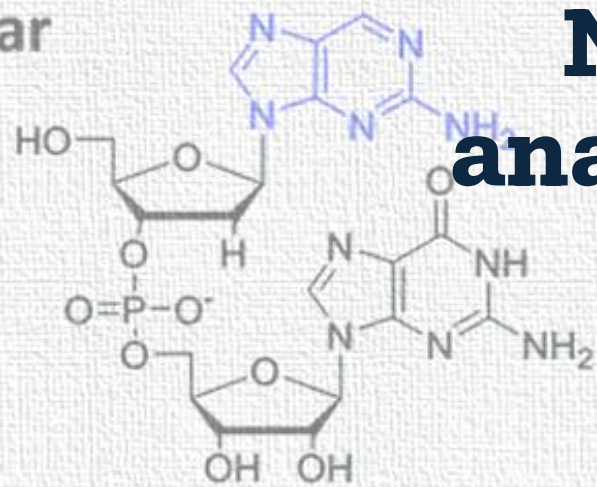
3'3'-cA(**etheno**)MP

4

ex: 345 nm
em: 435 nm



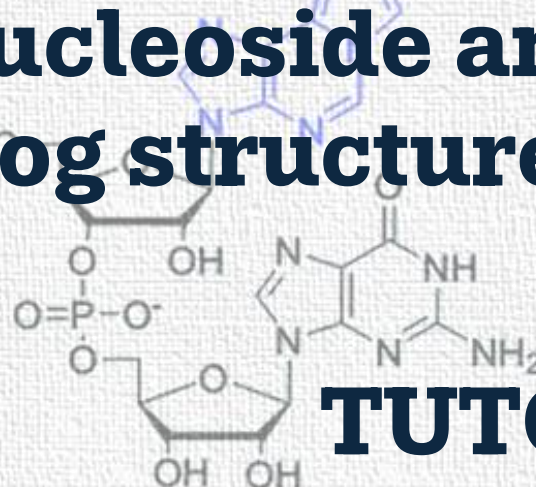
near



(**d2AP**)pG

5

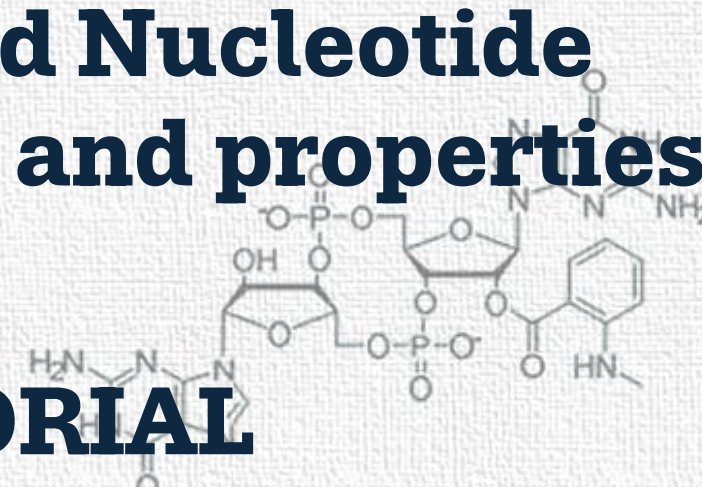
ex: 310 nm
em: 375 nm



(**etheno**)pG

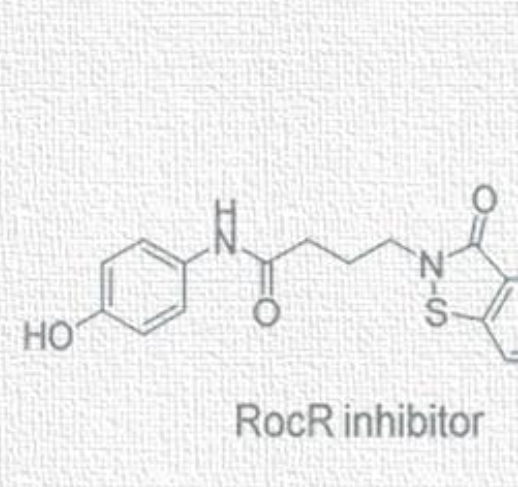
6

ex: 345 nm
em: 435 nm



MANT c-di-GMP

ex: 355 nm
em: 448 nm



RocR inhibitor

NucLigs: An online resource for Nucleoside and Nucleotide analog structure and properties

TUTORIAL

1

Open and access data

Access the database using the link nucligs.streamlit.app

Click on 'Launch database' to see the dataset and related information

2

Access the dataset and its related information through the interface

3

Apply filters and sorting

Keyword-based searches

Biological targets

Therapeutic application

Mode of action

NucLigs identifier

Substructure moiety

Bibliographic filter

Entire dataset search using the left side panel

4

Extract metadata file (.xlsx format)

For single analogs 'Extract selected' button

For range of analogs 'Extract range' button

For all filtered analogs 'Extract filtered' button



5

Generate 3D structures of analogs

Click on 'Generate PDBs' button



Run cell and load the extracted metadata file for selected analog(s)

Generated_PDBs.zip

Zip file with 3D coordinate structures and a generation report





NucLigs Database : A Nucleotide and Nucleoside Analog Database

[Tutorial](#)

NucLigs database

Nucleotide & Nucleoside Analog Database

11,007 compounds

6323 analogs

4658 natural

RDKit
2025.03.4

Theme: Dark

Upload Workbook

About NucLigs

Generate PDB



Search name, NL, SMILES, target, author, jourr



All Bases



All Types



All Regions



Bibliography Filters

11,007 results

Sort: Workbook order



↑

Show 100



Card View



Export Selected

Export Filtered

Range

1

to

50

Export Range

NucL_000001

N6-BENZYLADENOSINE

(2R,3R,4S,5R)-2-(6-(benzylamino)-9H-purin-9-yl)-5-(hydroxyme...

Adenine

Analog

2'-Deoxy

Sugar

357.40a

NucL_000002

6-amino-9-((2S,3R)-2-hydroxynonan-3-yl)-9H-purin-8-ol

Adenine

Natural

293.40a

NucL_000003

4-((2-chloro-9-isopropyl-9H-purin-6-yl)amino)benzoic acid

Adenine

Analog

Chloro

Base

331.80a

NucL_000004

(2S,3S,4R,5R)-5-(2-chloro-6-((3-chlorobenzyl)amino)-9H-purin-...

Adenine

Analog

Chloro

Base

483.40a

NucL_000005

(R)-9-(1-chlorooctan-2-yl)-9H-purin-6-a...

Adenine

Analog

Chloro

281.80a

NucL_000006

(2S,3S,4R,5R)-N-cyclopropyl-3,4-dihydrox...

Adenine

Natural

598.60a

NucL_000007

((E)-3-(6-amino-9-((2R,3R,4S,5R)-3,4-dih...

Adenine

Analog

2'-Deoxy

Sugar

414.40a

NucL_000008

(2R,3R,4S,5S)-2-(6-amino-2-(2-((E)-3-m...

Adenine

Natural

423.50a

NucL_000009

(5-(6-amino-9-neopentyl-9H-purin-8-yl)-...

Adenine

Natural

364.30a



Select a compound to explore

Search nucleotide and nucleoside analogs by IUPAC name, trade name, NL identifier, SMILES, or biological target.



Change theme

Upload workbook

Generate PDB structures using the exported metadata file

Understand about NucLigs database data statistics

Display panel for analog information

Search bar for keywords (target, substructure moiety, NucLigs ID, mode of action, etc)

Sorting data on the basis of nucleotide bases, type of analogs and region of modification

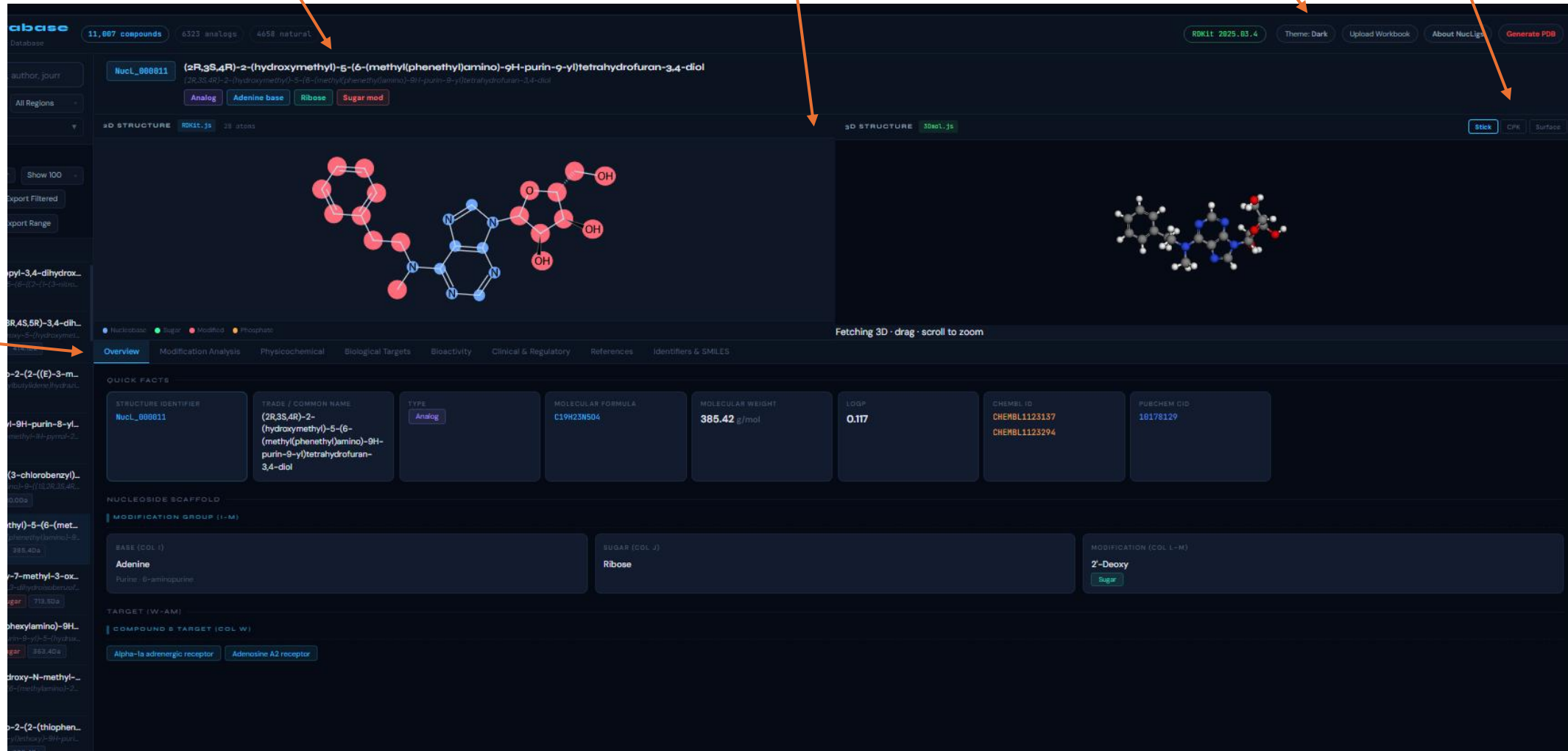
Rearrange the displayed workbook based on different parameters and show button

Export buttons on for selected analog/ filtered analogs or a range of displayed analogs

Analog entry with respective information



4658 natural



NucLigs_SMILES_to_3D.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

RAM Disk

Show code

NucLigs Database • Structure Generator

NucLigs 3D Structure Generator

Generate Desired 3D Structures in PDB Format

This professional module is a part of the **NucLigs Database** and enables automated generation of optimized **3D molecular structures in PDB format** from user-provided SMILES, CSV, or Excel files. Designed for computational biology, cheminformatics, RNA–ligand studies, docking, molecular dynamics, and structure-based drug discovery workflows. User need to upload the exported file from the NucLigs database in the cell and the tool will automatically generate the ZIP folder with all PDB format structures and an additional CSV report.

INPUT MODES
SMILES / CSV / XLSX

OUTPUT
Optimized 3D PDB

Generate 3D structures from resulted file

[1] Show code

37.1/37.1 MB 24.9 MB/s eta 0:00:00
4.9/4.9 MB 68.4 MB/s eta 0:00:00

Paste SMILES Upload File

Aspirin, CC(=O)OC1=CC=CC=C1C(=O)O
Caffeine, CN1C=NC2=C1C(=O)N(C(=O)N2C)C

OR

CCO
CCN

Paste SMILES → Click 'Generate PDBs'

Generate PDBs

User may either paste SMILES OR uploaded the extracted metadata file to generate 3D coordinate structures of desired analogs

Generate 3D structures from resulted file

[1] Show code

37.1/37.1 MB 24.9 MB/s eta 0:00:00
4.9/4.9 MB 68.4 MB/s eta 0:00:00

Paste SMILES Upload File

Upload CSV/XLSX

Automatically detects Name/NL and SMILES columns. Ignores other columns.

Upload (0)

Generate from File Delete File

Click on 'Upload'

Select the exported metadata file

Click on 'Generate from File'